"The expansive wave of the fifth long cycle in the United States: empirical evidence from the rate of profit"

Angélica Maza Albores[1] Seyka Verónica Sandoval Cabrera[2] Vladimir Márquez Stone[3]

Abstract

This work aims to verify the existence of an expansive wave corresponding to the fifth long cycle emerged in the United States after the 1973–1975 crisis. This study calculates the rate of profit, the rate of surplus value, and the organic composition of capital (OCC). The long-run movements are extracted with an Hodrick–Prescott filter and analyzed using a Vector Error Correction Model and Bai–Perron breakpoint tests. The results indicate the onset of a new expansive wave after 1983, driven in the first instance by a sharp increase in the rate of surplus value amid labor compensation reduction, and in the second instance by a slowdown in OCC growth associated with productivity gains from the fourth technological revolution.

JEL Classification: E320, E110, P1, N120, O510

Keywords: Fifth long cycle, United States, Rate of profit, Fourth technological revolution, Neoliberalism

[1]Facultad de Economía, Universidad Nacional Autónoma de México (UNAM). Email: angiemaza13@gmail.com. ORCID: 0000-0003-1552-9823.

[2]Facultad de Economía, Universidad Nacional Autónoma de México (UNAM). Email: seykasandoval@comunidad.unam.mx. ORCID: 0000-0003-1552-9823.

[3]Facultad de Economía, Universidad Nacional Autónoma de México (UNAM). Email: vladimirmarquezst@comunidad.unam.mx. ORCID: 0009-0001-1187-6580.

1. Introduction

The 1973–1975 crisis in the United States set in motion a cluster of economic, productive, and social changes in the world economy that deteriorated the historical stage of Fordist-Keynesian capitalist development (Harvey 2007: 129; Ordóñez 2024). In terms of long-term oscillations (waves, cycles), the crisis corresponds to the depressive wave of the fourth long cycle whose expansive wave covered the period 1948–1967 and its depressive wave began in 1968 (Mandel 1995: 82).

The literature review[4] agrees that the crisis of 1973–1975 buried profitability that had been slowing down since 1960, forming a depressive wave (Li et al. 2007; Maito 2013; Basu 2016; Duménil and Lévy, 2016; Shaikh 2016; Tsoulfidis and Tsaliki 2022). However, there is no consensus among scholars on the existence of the fifth long cycle. Some authors argue that advanced countries were still in the depressive wave of the fifth cycle (Mandel 1995), while others propose the hypothesis of the beginning of an expansive long wave of the fifth Kondratieff cycle in the 1980s (Li et al. 2007; Maito 2013; Basu 2016; Duménil and Lévy 2016; Shaikh 2016; Tsoulfidis and Tsaliki 2022

Based on Ernest Mandel's theoretical framework and his periodization of the fourth long cycle, this paper aims to verify the existence, and the cause of the expansive wave of the fifth long cycle specifically in the United States, a country that we consider to be the current hegemon of the capitalist mode of production[5] due to its contribution to the world GDP and its leadership in the

[4] We followed a semi-systematic literature review (Snyder 2019).

[5] Following the criteria of Marx (1982: 90), who analyzed the most developed country at the time, England. We note a weakening of the hegemony of the United States, although, given the contribution to GDP and the technological leadership of the 4th technological revolution, we consider the characterization as a hegemonic country is sustained. In 2021, the United States was the largest contributor to the world's real GDP (at 2015 US dollars) with 23.5 percent, followed by China, which contributed 18.2 percent (World Bank 2024). In the contribution to world GDP measured in Purchasing Power Parity (PPP) at 2021 US dollars, China ranked first with 18.6 percent, and the United States in second place with 15 percent (World Bank 2024a). The United States is the world leader in patents development for digital production technologies, and in their implementation in the manufacturing sector (United Nations Industrial Development Organization 2019: 5).

digital production of the 4th technological revolution.[6] To this end, the key variables of the rate of profit are calculated for the empirical verification of long waves: the organic composition of capital (OCC) and the rate of surplus value.

To estimate the long-term movements of the profit rate in the United States, a methodology grounded in Marxian categories was employed, following the approach of Shaikh and Tonak (2007), which derives surplus value, constant capital, and variable capital from input-output tables reorganized by productive and unproductive sectors according to the North American Industry Classification System (NAICS). The distinction drawn by the authors is further refined using the sectoral distinctions proposed by Mandel (1992). The Hodrick–Prescott filter was applied to the rate of profit series to identify the long-cycle trend; a cointegration model with a vector error correction mechanism (VECM) was employed to analyze the profit rate series; and the Bai–Perron structural break test was used to determine the years of structural change in the US economy.

The findings indicate that a new long wave began in the United States after 1983, marking the onset of the fifth long cycle, when the rate of profit experienced a sharp upturn—a level shift exceeding its trend in the previous decade. The causes of this inflection are, in the first instance, the increase in the rate of surplus value thanks to the reduction of labor compensation; and in the second instance, the slowdown in the rate of increase of the OCC derived from the recovery in labor productivity due to the massive application of the 4th technological revolution.

The paper is organized into four sections. The first is the methodological proposal for calculating the rate of profit, the rate of surplus value, and the organic composition of capital (OCC), and the

[6] Based on the methodological criteria for identifying a technological revolution according to Mandel (1995: 33–34), we consider these technological changes that occurred in the 1970s as the prelude to the 4th technological revolution. The first phase of deployment of this technical change happened in the early 1980s when capitalist profitability was recovered, which led to investment in "the first massive applications of these basic innovations", once the inflection from a long depressive wave to a long expansive wave had already occurred (Mandel 1995: 30).

econometric techniques employed for the analysis of the profit rate and its components; the second section presents the empirical results to verify the existence of a new expansionary long wave, and the third discusses these results within the framework of the literature review. The paper concludes with general findings.

2. Methodology

The literature review on long cycles in the United States presents various methodologies for empirical corroboration. All of them contain estimates of the rate of profit. Li et al. (2007), Maito (2013), and Roberts (2016) use indicators for the entire economy. Shaikh (2016) and Tsoulfidis and Tsaliki (2022) use corporate profits. Basu (2016) considers the share of profit income in nominal net domestic product. We measured the profit rate for productive activities.

Shaikh and Tonak (2007) offer a methodology for calculating the rate of profit, specifically for productive activities. They start with input-output (IO) tables and, based on monetary flows, derive the Marxian categories for its measurement. As is well known, the columns and rows of the IO tables are organized according to the North American Industry Classification System (NAICS), therefore, to distinguish between productive and unproductive activities, the authors divide the IO table into three sectoral categories: production, trade, royalties, and dummy sectors, which, together with final demand, the sum of the columns equals the gross output, and the sum of the rows is the gross production (Shaikh and Tonak 2007: 74–75). The authors consider only the production and trade sectors as productive activities (Shaikh and Tonak 2007: 40). The transformation of IO measures into Marxian measures for the rate of profit is summarized in table 1 below.

Marxian measure	Input–Output measure
S* = Surplus value	
S* = VA*–V*	
VA* = Marxian value added	
VA* = VAp + VAt + RYp + RYt + (Mp)t + (Mt)t	VAp = Value added of the production sector VAt = Value added of the trade sector RYp + RYt = Royalties of the production and trade sectors (Mp)t + (Mt)t = trading margin on inputs in the production and trade sectors
Mp = constant capital used up (flow)	
K* = Stock of capital	Fixed nonresidential gross private capital
V* = Wp = Wage bill of the productive workers = Variable capital	Compensation of productive full-time, part-time and self-employed persons
K*/V* = Value composition of capital	
S*/V* = rate of surplus value	
r*= S* / K* = Marxian general rate of profit	

Note: The trading margin is the difference between the producer's price and purchaser's price (Shaikh and Tonak 2007: 45).

Source: Shaikh and Tonak (2007).

Table 1. Correspondence between Input–Output Measures and Marxian Measures for the Calculation of the Rate of Profit

For Shaikh and Tonak (2007: 108–109), productive labor is found in the following NAICS sectors: "agriculture, mining, construction, transportation and public utilities, manufacturing, and productive services (defined as all services except business services, legal services, and private households)". Part of the productive services include wholesale /retail trade; rental of produced commodities such as cars, buildings, and equipment; government trading enterprises, and transportation for the realization of these activities (Shaikh and Tonak 2007: 40). We consider that these activities do not constitute productive labor; rather, what is "generated" in those sectors represents a distribution of surplus value produced in the truly productive sectors.

Therefore, we adhere to Mandel's definition of productive labor. For the author, labor is productive when the commodities it produces enable the reproduction and expansion of global capital; that is, "labor increases the global mass of surplus value" (Mandel 1992: 41–42). The use-value of unproductive labor is not central from the global capital perspective, since even if a specific kind of labor may be useful and necessary for society in general, or even for the individual capitalist, as long as it does not add value—and consequently surplus value—but merely redistributes already created value, such labor is considered unproductive.

Accordingly, the author lists several specific types of work that fall under the category of productive labor: transportation of goods; manufacturing and mining industries; public utility services ("production and transportation of water or any form of energy," such as gas and electricity); construction and sale of houses and offices; provision of construction materials; and services such as food sales in restaurants, film production, engineers, laboratory scientists, supervisors, managers, and warehouse employees (Mandel 1992: 44–46).

Thus, for the rate of profit, we consider the following indicators (in current prices) from the US National Income and Product Accounts (NIPAs):

- VA* = Marxian value added = VAp+Mp
- VAp = Gross value added of the productive sectors
- Mp = Constant capital used up (flow) = Gross domestic investment
- V* = Compensation of full- and part-time employees
- S* = Surplus value = VA*−V*
- K* = Stock of capital = Net stock of private nonresidential fixed assets at current cost
- K*/V* = Organic composition of capital

- S*/V* = Rate of surplus value
- r* = S*/C* = Rate of profit

Considering the following sectors and subsectors of the US economy: Agriculture, forestry, fishing, and hunting; Mining; Utilities; Construction; Manufacturing; Transportation and warehousing (except Transit and ground passenger transportation); information; Professional, scientific, and technical services (only Computer systems design and related services, and Miscellaneous professional, scientific, and technical services); Management of companies and enterprises, and Food services and drinking places).

Then, using an effective sample size of 50 observations (1974–2023), the smooth trend of the rate of profit series was extracted using a Hodrick–Prescott filter, with $\lambda = 100$, which is appropriate for annual data. The result is presented in figure 1. The analysis of the rate of profit and its components was conducted using a vector error-correction model (VECM). This model is underpinned by the results of the Johansen cointegration test, which indicated the existence of at least one cointegration relationship among the three variables (see appendix). This justified the use of a VECM to capture the system's short and long-run dynamics. Also, the Bai–Perron multiple breakpoint test was used to examine structural shifts in the relationship among the rate of profit, the rate of surplus value, and the organic composition of capital.[7] Next section begins with key indicators employed by Mandel to verify the existence of long waves.

[7] The study acknowledges that its approach does not seek to estimate the full periodicity of multiple Kondratieff cycles or to construct a comprehensive chronology of long waves for the US economy. Instead, it adopts econometric techniques suited to the sample size and is conceived as a case study that empirically verifies the existence of a prolonged recovery in profitability during the 1980s.

3. Results

According to Mandel (1976: 115), following Marx, labor productivity is one of the fundamental variables for empirically verifying the existence of long waves, as it affects the mass of surplus value produced and the increase in the rate of surplus value—both elements that determine the rate of profit. Other variables from both the productive sphere and the sphere of circulation are also considered fundamental by Mandel (1976) for verifying long waves: industrial production and world trade per capita. The former reflects "the long-term tendency of capitalist *production* and the latter the rhythm of expansion of the *world market*" (Mandel 1976: 141–142).

In terms of the causes of long waves, these variables would reflect the endogenous cause of technological progress, which sustains commodity production and extraordinary profits, and the exogenous factor of the geographical expansion of capitalist production and the change in competitive relations according to participation in the world market—both determinants for the recovery of the rate of profit (Mandel 1995: 17; Marx 1991: 364).

Table 2 shows the variation rates of indicators corresponding to the main variables identified by Mandel, considering the availability of historical data series as well. For measuring labor productivity, GDP per capita is appropriate, since Maddison Project Database provides a continuous time series for the US economy since the 1800s. Variables related to prices, and the interest rate were also included.

Long cycle	Wave	Period	GDP per Capita	GNP per Capita	Index of Manufacturing Production	60-day Commercial Paper Interest Rate	Exports per Capita	General Price Level	Long-Term Government Bond Yields
			Average growth rate				Compound growth rate		
First	Expansive	1789-1825	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Depressive	1826-1848	1.1	N/A	N/A	N/A	N/A	N/A	N/A
Second	Expansive	1848-1873	1.4	N/A	N/A	N/A	N/A	N/A	N/A
	Depressive	1874-1893	1.2**	n/a	4.5	5.2	0.1	-1.2	N/A
Third	Expansive	1894-1913	2.1	2.3	5.7	4.6	3.8	1.7	N/A
	Depressive	1914-1939	1.3	1***	3.2	3.6	0.5	1.6	N/A
Fourth	Expansive	1940-1966	2.7	3.2	6.3	2.4	6.1	3.2	2.8
	Depressive	1967-1982	1.6	1.6	2.3	8.1	12.7	6.3	3.3
Fifth	Expansive	1983-2006	2.3	2.4	3.5		8.3	4.4	-3.1
	Depressive	2007-2023	0.9*	1.2*	-0.2		3	2.5	-0.5

Notes: N/A not available. * Up to available data for 2022. ** It considers 1874–1894 as the original period. *** The year 1939 is excluded because its level was 11 times higher than in 1938, resulting in an annual growth rate of 41%.

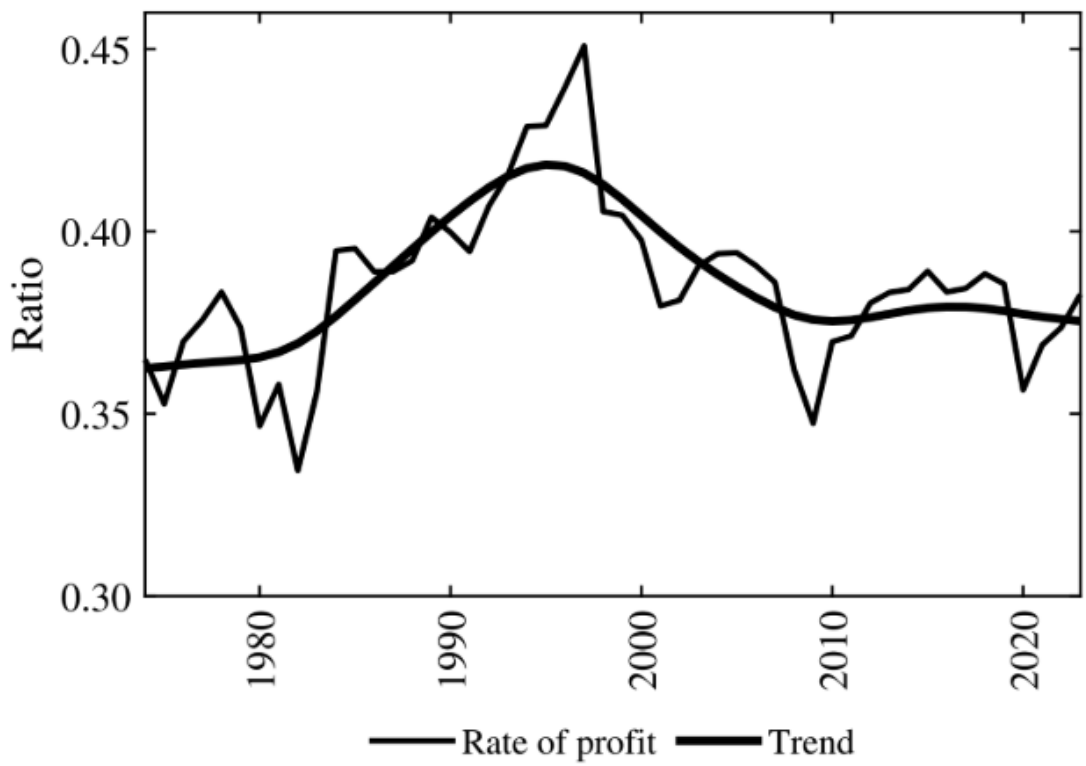
Sources: Board of Governors of the Federal Reserve System (2025, 2025b), Bolt and van Zanden (2024), National Bureau of Economic Research (2025a, 2025a), Organization for Economic Co-operation and Development (2025), Smits et al. (2009), and U.S. Bureau of Economic Analysis (2025).

Table 2. Evolution of Long Cycles: Expansive and Depressive Waves with Selected Economic Indicators (1789–2023)

Unlike the other indicators, which show higher variation rates during expansive waves than in depressive ones, this pattern does not hold for exports per capita or the commercial paper interest rates. During the depressive wave of the fourth long cycle in the United States, exports per capita were higher than during the expansive wave. This is due to the increase in exports driven by rising oil prices following the 1979 energy crisis, which extended until 1981. That same crisis triggered an acceleration of inflation and a rise in interest rates, because of the restrictive monetary policy implemented under the chairmanship of Paul Volcker at the Federal Reserve, which negatively affected stock market performance (Shiller 2015). That is the reason behind the 60-Day Commercial Paper Interest Rate in the depressive wave of the fourth cycle.

The trend curve of the rate of profit is shown in figure 1. The profit rate of the US economy peaked in 1997, with troughs in the mid-1980s and in 2009. The expansionary wave begins in 1984, when the rate of profit rose sharply after a V-shaped recovery after a depressive trend during the previous decade. This growth continues until it peaks in 1997. In the following year, the rate of profit entered

a downward trend, reaching its lowest point during the 2007–2009 crisis. Despite the V-shaped recovery, it did not exceed the peak of the 1990s and even tended toward stagnation. Its effects have been more prolonged, as they did not produce a radical trend reversal as in 1983, but instead kept profitability within the same range as in the inter-crisis period.



Note: Trend estimated with the Hodrick–Prescott filter (λ = 100).
Sources: U.S. Bureau of Economic Analysis (2025a, 2025b, 2025c, 2025d).

Figure 1. Evolution of the Profit Rate and its Trend in the United States (1974–2023)

Theoretically, the organic composition of capital (OCC) and the rate of surplus value determine the movements of the profit rate. A cointegration equation was estimated through a VECM to analyze the relationships among these variables for the 1974–2023 period. Prior to estimating the VECM, the optimal lag order was determined using standard information criteria, as reported in the appendix. Accordingly, the VECM was estimated with rank r=1 and lag order p=1. To

establish the order of integration of the series, we conducted the Kwiatkowski–Phillips–Schmidt –Shin (KPSS) and the Augmented Dickey–Fuller (ADF) tests for the profit rate, surplus value, and OCC. The combined results are presented in the appendix.

For OCC and surplus value, the null of stationarity is rejected by KPSS, while the ADF fails to reject the unit root null, which is consistent with treating both series as I(1) processes. For the profit rate, the KPSS test does not reject the level-stationarity null, while the ADF likewise fails to reject the unit root. This conflicting evidence leads us to classify the profit rate as borderline and "inconclusive" in levels. In practice, however, all three variables become stationary once first differences are taken, which supports their treatment as integrated of order one, I(1). Therefore, in what follows, we proceed under the assumption that OCC, surplus value, and the profit rate are I(1), and may cointegrate in the long run.

In table 3, Panel A reports Johansen trace statistics under a specification with no trend and a constant restricted to the cointegration space. At the 5 percent level, the null of no cointegration (r=0) is rejected, while the null r≤1 cannot be rejected. This implies the presence of at most one cointegrating relation, so we proceed with r=1 as the baseline rank. Panel B reports the cointegrating vector normalized on OCC, which implies that, in equilibrium, a higher rate of surplus value (*Surplus value*) is associated with a lower *OCC*, while a higher rate of profit (*Profit*) is positively related. The estimated long-run relation can be expressed as:

$$\mathrm{OCC}_{t-1} - 2.648\ \text{Surplus value}_{t-1} + 19.750\ \mathrm{Profit}_{t-1} - 6.466 = 0$$

Panel C reports the loading (error–correction) coefficients. The adjustment terms are $\alpha_{\Delta \mathrm{OCC}} = 3.230$, $\alpha_{\Delta \text{Surplus value}} = 1.221$, and $\alpha_{\Delta \mathrm{Profit}} = -0.031$. The magnitudes suggest that OCC responds strongly and positively to equilibrium deviations; the rate of surplus value also corrects, though

with less intensity; and the rate of profit rate practically does not adjust since the coefficient is close to zero.

Panel A: Trace Test Statistics				
Null hypothesis	Test statistic	10% crit.	5% crit.	1% crit.
$r \leq 2$	4.11	7.52	9.24	12.97
$r \leq 1$	14.82	17.85	19.96	24.6
$r = 0$	37.89	32	34.91	41.07
Decision: rank r = 1 (5% level)				

Panel B: Cointegration Vector (normalized on OCC)	
Variable	Coefficient
OCC_{t-1}	1.000
Surplus $value_{t-1}$	−2.648
$Profit_{t-1}$	19.75
Constant	−6.466

Panel C: Adjustment Coefficients (Loading Matrix α)			
	OCC Δ	Surplus value Δ	Profit Δ
Coefficient	3.23	1.221	−0.031

Source: Authors' calculations using the urca package in R.

Table 3. Johansen Cointegration Results

Table 4 reports the short-run dynamics and the error-correction term from the VECM with rank r=1 and lag order p=1. Among the short-run dynamics, lagged changes in OCC have a positive effect on the rate of surplus value (+0.778) and the rate of profit (+0.135), although of smaller magnitude. Lagged changes in the rate of surplus value have a strong positive effect on OCC (+1.874), a negative effect on itself (−1.450), and a slight negative effect on the rate of profit (−0.320). Finally, lagged changes in the rate of profit.

	ΔOCC	ΔSurplusValue	ΔProfit
ECT_{t-1}	3.23	1.221	−0.031
ΔOCC_{t-1}	−0.392	0.778	0.135
$\Delta Surplus\ value_{t-1}$	1.874	−1.45	−0.32
$\Delta Profit_{t-1}$	−16.791	9.703	2.376

Note: Values rounded to three decimals.

Source: Authors' calculations using the urca package in R.

Table 4. VECM Short-run Dynamics, and Error-correction Term

Table 5 presents the Bai-Perron multiple breakpoint test results for the relationship between the rate of profit, the rate of surplus value, and the OCC. This indicates the presence of three structural breaks, yielding a four-segment partition of the sample. Breaks are estimated around 1985, 2007, and 2021, with 95 percent confidence intervals covering 1984–1986, 2006–2008, and 2020–2022, respectively. These results suggest that the profit-surplus value-OCC relationship is not stable over the full period 1974–2023, but instead exhibits distinct regimes separated by three structural changes.

	Breakpoint 1	Breakpoint 2	Breakpoint 3
Estimated year	1985	2007	2021
95% CI (years)	[1984, 1986]	[2006, 2008]	[2020, 2022]

Source: Authors' calculations using the urca package in R.

Table 5. Bai-Perron Structural Break Test Results (Years)

The results presented in this section will be discussed in the next one considering the study's hypothesis.

4. Discussion

One of the most relevant aspects in the study of long waves is their periodization. Based on the authors reviewed in the literature, table 6 shows the identified peaks and troughs for the US

economy. Using Mandel's periodization as a reference, for the fourth long cycle (approximately from the postwar period to the 1980s), there is broad consensus in locating the peak in the mid-1960s—1963 (Tsoulfidis and Tsaliki 2022), 1965 (Roberts 2017), 1966 (Maito 2013, and this research), and 1967 (Mandel 1995; Basu 2016).

The trough is in the early to mid-1980s, particularly 1982, as identified by most authors, and therefore, the beginning of a new expansionary long wave. This research is consistent with those authors, based on the results in table 5 that suggest a breakpoint between 1984 and 1986. As shown in figure 1, 1982 marks the trough of the rate of profit. The V-shaped recovery is achieved in 1983, allowing us to interpret that the expansionary phase of the current fifth long cycle began in 1984.

Regarding the fifth long cycle, authors agree on a peak around the late 1990s or the first decade of the XXI century: 1997 (Maito 2013, and this study),[8] 1998 (Li et al. 2007), 2000 (Shaikh 2013), 2002 (Roberts 2017), 2004 (Duménil and Lévy 2016), and 2007–2008 (Basu 2016; Tsoulfidis and Tsaliki 2022). The results in table 5 indicate that the year marking the structural change—the beginning of a new trend relative to the previous one—was 2009.

While some authors from table 6 propose the US mortgage crisis as the peak of the expansionary wave, the results of the previous section instead suggest the trough of the long cycle right in the midst of the crisis.[9] These results also advise that the US economy has been in a depressive phase since then. Nonetheless, Duménil and Lévy (2016), Roberts (2016), and Tsoulfidis and Tsaliki

[8] In 1997, as can be interpreted from table 4, a crisis was in progress since an increase in the rate of surplus value drove a decline in the rate of profit

[9] The breakpoints from table 5 provide support for the qualitative differentiation of capitalist crises: while some may be longer or deeper, others of a structural nature mark the onset of new dynamics in capital accumulation. In terms of the rate of profit, the 1980 crisis in the United States was deeper than that of 1973–1975, just as the 2007–2009 crisis was deeper than that of 2001. Nevertheless, in both cases, the structural crises reflected a shift in trend already reached in previous years: in 1966 and in 1997, respectively.

(2022) also propose projections of the end of the depressive wave: Duménil and Lévy (2016) in 2014, 2018 for Roberts and Shaikh, and up to 2028 for Tsoulfidis and Tsaliki.

Author	Peak and trough	Duration of the expansive wave (years)	Temporal coverage	Database (variables)	Methodology employed
This study	Peak: 1966 / Trough: 1982 / Peak: 1997 / Trough: 2009	16 years	1940–2023	BEA, BLS, FRED (rate of profit, OCC, rate of surplus value)	Hodrick-Prescott filter (lambda =100)
Mandel (1995)	Trough: 1940 / Peak: 1967 / Trough: ¿2000?	28 years	1789–1968	Van Duijin (1979), Devleeshouwer (1970), Imbert (1956) (world trade, industrial output, index per capita world production, output of energy)	Annual rate of growth
Li et al. (2007)	Trough: 1939 / Peak: 1950 / Peak: 1969 / Trough: 1983 / Peak: 1998 / Trough: 2005?	16 years	1869–2005	BEA, Census Bureau, Duménil and Levy (1993) (profit rate, rate of accumulation)	10-year moving average
Maito (2013)	Peak: 1966 / Trough: 1982 / Peak: 1997	16 years	1960–2009	BEA, BLS (profit rate adjusted for capital turnover), annual rate of surplus value, capital value composition)	Observational inference
Shaikh (2013)	Trough: 1938 / Peak: 1963 / Trough: 1984 / Peak: 2000 / Trough: 2018	17 years	1890–2009	Measuring Worth.com (U.S. Gold price)	Hodrick-Prescott filter (lambda =100)
Basu (2016)	Trough: 1941 / Peak: 1967 Trough: 1982 / Peak: 2007	26 years	1848-2010	Maddison Project (real gross domestic product, real per capita GDP)	Annual rate of growth
Duménil and Lévy (2016)	Trough: 1910 / Peak: 1963 / Trough: 1986 / Peak: 2004 / Trough: 2014	19 years	1870–2014	BEA (profit rate, productivity of capital)	Hodrick-Prescott filter
Roberts (2017)	Trough: 1946 Peak: 1965 / Trough: 1982 / Peak: 2002 / Trough: ~2018	21 years	1950-2014	Penn World Tables (rate of profit, rate of surplus value, OCC)	Annual rate of growth
Tsoulfidis and Tsaliki (2022)	Trough: 1947 / Peak: 1963 / Trough: 1982 / Peak: 2007-2008/ Trough: ~2028	26 years	1940–2020	FRED (profit rate, mass of real net profit)	Nonlinear logistic adjustment

Sources: Basu (2016: 32), Duménil and Lévy (2016: 532), Li et al. (2007:40), Maito (2013: 14), Mandel (1995: 82), Shaikh (2013: 749), Roberts (2017: 228), and Tsoulfidis and Tsaliki (2022: 15).

Table 6. Long Cycles Studies: Peaks, Troughs, Duration, Databases, and Methodologies

About the duration of the last expansionary wave, this study closely matches Maito (2013) and Li et al. (2007), reinforcing the existence of a ~16-year cycle. Studies such as Tsoulfidis and Tsaliki (2022) or Basu (2016) show longer waves (≈26 years), reflecting methodological and theoretical differences: while some use statistical filters (Hodrick–Prescott filter, moving averages), while others apply nonlinear logistic fits or growth rates. Overall, this study is consistent with all authors in placing the beginning of an expansionary wave in the early to mid-1980s, and for most scholars, a new peak in the late 1990s. Additionally, table 5 identifies the crisis triggered by the COVID-19 pandemic as another breakpoint. This phenomenon is not addressed in the present study; therefore, works such as that of Tsoulfidis and Tsaliki (2022) are required to consider the most recent developments and their impact on the long-term profit rate.

Although all the authors reviewed belong to the Marxist tradition, they differ in their interpretation of the causes underlying the expansionary wave of the 1980s. The endogenous causes—economic

and technical in nature—influence and are shaped by "the longer-term evolution of the productive forces" (Louçã 1999: 188); while exogenous causes focus on factors external to the general process of socioeconomic development, also referred to as extra-economic factors (Louçã 1999).

This study seeks to recover and highlight Ernest Mandel's contributions for the analysis of a specific case: the US economy. Mandel argued that a Marxist analysis of long waves must be based on the long-term movements of the rate of profit, which determine, "in the last analysis, quicker and slower long-term paces in capital accumulation" (Mandel 1995: VII). A sharp increase in the profit rate is the key indicator of a new expansive wave, which occurs when several or all of the following elements operate in a synchronized manner, overcoming its tendential fall: sharp increases in the rate of surplus value, a slowdown in the organic composition of capital growth, acceleration of capital circulation, an increase in the mass of surplus value, and the relocation of capital to countries and/or sectors with an organic composition of capital below the average of industrial sectors in advanced economies (Mandel 1995: 11).

Hence, these five determinants of the rate of profit rate ranked by the author according to the long wave in which the capitalist system is situated: endogenous factors drive the transition from boom to depression—especially the rise in the organic composition of capital and in labor productivity —while exogenous factors are necessary to exit the depressive phase, particularly through the emergence of a new technological revolution (Mandel 1981: 332).

Among the exogenous causes are wars, revolutions and counterrevolutions, the geographical expansion of capital, and—most centrally—changes in the balance of power between social classes (Mandel 1995: 16–17). In particular, the defeat of the working class at key historical moments allows for a sharp rise in the rate of surplus value, which can reverse the downward trend in

profitability and trigger a new expansive wave in the long cycle (Mandel 1981: 336). This political-social dimension is key to understanding why the transition from a depressive wave to an expansive long wave is not purely endogenous but rather reflects a multicausal perspective. Since subjective factors are unpredictable and historically unrepeatable there can be no fixed periodicity (such as 50–60 years) or symmetry in the duration of the waves (Mandel 1995: 16–17).

Once the ascending phase begins, the growth of the profit rate can be sustained through the diffusion of a new technological revolution. These revolutions do not arise spontaneously but require prior conditions that emerge during depressive waves: accumulated idle capital, available but not yet disseminated inventions, and a structural need to renew the productive apparatus to restore profitability (Mandel 1995: 30–31). In this framework, the technological revolution consists of massive investment in innovations that enables a radical transformation of production processes, transport and telecommunications systems, and the sectoral structure of the economy.

Within this dynamic, it is crucial to distinguish between invention and innovation. While invention refers to the technical creation of new knowledge or discoveries, innovation corresponds to their incorporation into the capitalist valorization process. A true expansive long wave is only triggered when innovation succeeds in renewing fixed capital on a large scale—particularly in Department I, which produces means of production (Mandel 1976: 119–120). The author proposes a ten-year period between the intensification of research and basic innovations and investment in the first large-scale applications of these innovations. This corresponds to the first phase of a technological revolution, once the inflection point from a depressive long wave to an expansive one has occurred (Mandel 1995: 44–45).

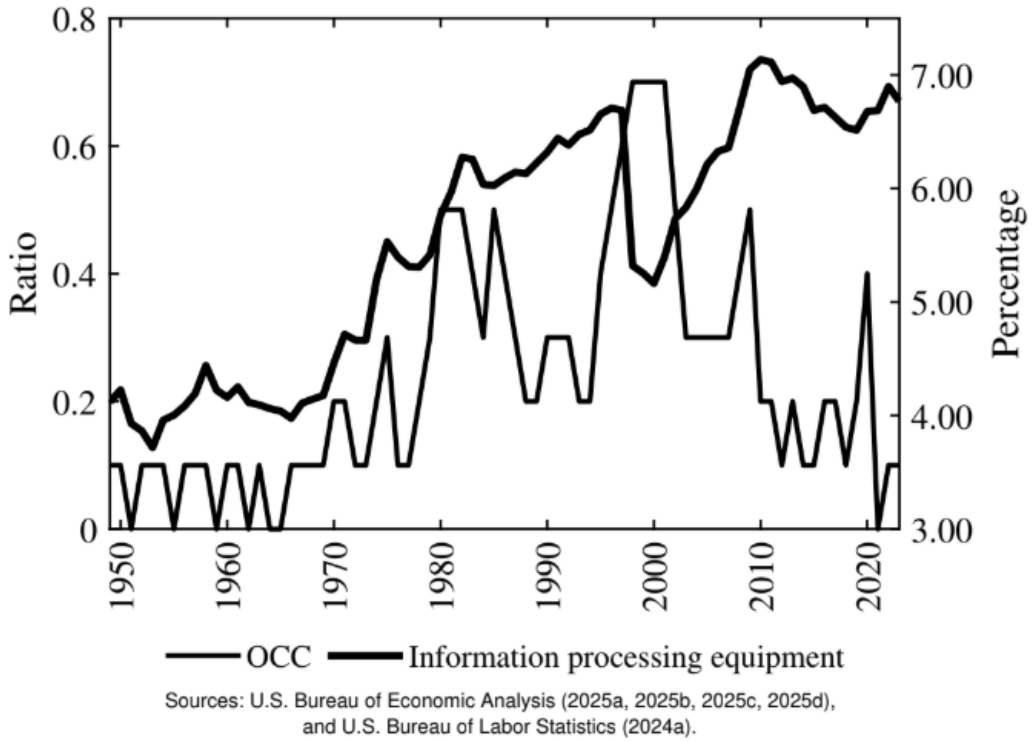
Long waves are not merely expressions of the cyclical movement of key indicators from an economy—which would be devoid of substantive meaning on its own—but rather, in a fuller sense, they constitute historical stages or phases in the development of capitalism. These are structured into ascending and descending waves which, when combined, form long cycles (Mandel 1986: 92). In this way, historical phases and long waves are integrated into the explanation of the development of the capitalist mode of production, driven by the long-term movement of accumulation and the rate of profit.

In our case study, the expansive wave corresponds to a new phase of development which replaced Keynesian Fordism. The informatic capitalism was driven by the information and telecommunications revolution and based on knowledge and information as the main productive power (Castells 2022; Ordoñez 2024). This 4th technological revolution has computer equipment and telecommunication networks as the fundamental instruments of labor for the accumulation of capital. The set of branches that produce these instruments is the informatics-electronics sector (Ordoñez 2023).

The following figure 2 shows the growth of the contribution of information processing equipment to labor productivity[10], and therefore to the production of relative surplus value, exhibiting a change in trend in 1970. The contribution of the main instruments of labor to productivity surged after the 1973–1975 crisis, in the context of competition between capitalists who faced a devaluation of their capital resulting from this crisis. At the same time, OCC was showing a near-exponential rise in the 1970s, with corresponds to a period of depressed profitability. Considering the main instrument of labor, and the OCC, the 1970's reflect "the flourishing of groundbreaking

[10] The contribution of information processing equipment (ipe) intensity to labor productivity is "the portion of labor productivity change attributed to the change in the use of information processing equipment capital inputs relative to hours worked. Information processing equipment capital includes asset types such as computers and communications equipment" (U.S. Bureau of Labor Statistics 2024b).

innovations" during a long-lasting recession and declining profitability (Tsoulfidis and Tsaliki 2022: 6).



Figure

2. Growth Rate of Information Processing Equipment's Contribution to Labor Productivity, and OCC (1949–2023)

This is consistent with Tsoulfidis and Tsaliki (2022: 11), who pointed out that "disruptive innovations" are introduced towards the end of the depressive wave of the long cycle. Therefore, 1970's is interpreted as the beginning of the productive adoption of the technologies of the 4th revolution through the intensification of research and initial basic innovations as the first attempt by capital to increase the rate of surplus value (Mandel 1995: 34). However, the leap in the growth of the contribution of the means of labor does not occur until the beginning of 1980, when the rate of surplus value and the rate of profit changed trends.

According to table 4, past increases in the rate of surplus value led to a short-run expansion of the OCC; and in the long run, higher levels of rate of profits are associated with an increase of OCC, as is shown in table 3 (Panel B). That is, a recovery of profitability was required for the first phase of massive use of the new technology at a national level, changing the trend of labor productivity (output per hour), as shown in figure 3. Following Mandel (1995: 30, 44), it took ten years for the capital idly hoarded during the crises of 1973–1975 and 1980 to be invested in the massive applications of these basic innovations of the 4th technological revolution once the general conditions of the social production process were profitable.[11]

Between 1967 and 1983, labor productivity followed a downward trend, with a negative slope and an average growth rate of 1.9percent. In contrast, from 1984 to 2009, productivity recovered, averaging 2.4percent annual growth. In other words, technological change (reflected in the increase of the OCC during the 1970s) was a condition for raising labor productivity. This upturn will lead to the cheapening of the elements of constant capital, and therefore, the slow growth of the OCC in the next decade, which, with the increase in the rate of surplus value, generated a rise in the rate of profit in the 1980s.

[11] Global technological revolutions are described by Mandel (1976: 116) as "phases of the re-entry of idle capital into the process of valorization, determined by a sudden rise in the average rate of profit".

Note: Trend estimated with the Hodrick–Prescott filter (λ = 100).
Source: U.S. Bureau of Labor Statistics (2024).

Figure 3. Growth Rate of the Labor Productivity (1967–2023)

The result above aligns with consistent with Maito (2013: 200) whose empirical results show that during the first half of 1980 in the United States, the rate of surplus value grew more than the OCC, doubling capitalist profitability compared to the previous five years. The results of Roberts (2016: 224) are similar, as he shows that in the period 1982–2002, the conjunction between the negative growth of the OCC (−5.2) and a rate of surplus value of 22.5 percent resulted in a rate of profit of 29.9 percent. However, what we consider the beginning of the expansionary long wave of the fifth long cycle in mid-1980s, for Roberts, (2016: 228–230) is a new profit cycle in its season of rising profitability within the downward wave of the IV Kondratieff cycle.[12]

[12] "[…] the profit cycle integrates with the K-cycle into what [I] have been called four seasons: spring (rising profitability); summer (falling profitability) alongside the rising phase of the K-cycle; and then autumn (rising profitability) and winter (falling profitability), alongside the declining phase of K-cycle" (Roberts 2016: 230).

However, the underlying cause of the upturn in the rate of surplus value has not yet been examined. Table 4 shows that an increase in the rate of profit drives a decrease in the OCC. That is, in theoretical terms, surplus value may stem from an extension of the working day or from wages below their social value. The first restructuring factor, typical of crises to upsurge the rate of surplus value according to the Marxist approach, is unemployment since it implies a reduction in variable capital. The unemployment rate went from 4.9 percent in 1973 to 9.7 percent in 1982 (U.S. Bureau of Labor Statistics 2023c), which explains the increase in the OCC while its rate was declining in those years. However, unemployment was not enough for the recovery of profitability since the maximum gap experienced between the rate of surplus value and the OCC in the United States occurred until the 1980s, according to Maito (2013: 200) and Roberts (2016: 225).

The latter is confirmed by Deepankar's Basu analysis of recessions in US business cycles. For Basu (2016: 5), business cycles are characterized as reproductive cycles, i.e., the increase in the Industrial Reserve Army (IRA) restores profitability expectations, and non-reproductive cycles, when changes in the IRA fail to restore such expectations. According to Basu (2016: 36), the 1973-1975 crisis in the United States was a non-reproductive cycle, while the 1980 recession was a reproductive cycle. Hence, the introduction of the 4th technological revolution remained an attempt to reverse the decline in the rate of profit since the 1960s.

The recovery of the rate of profit required an additional factor to unemployment and the devaluation of capital. As Mandel (1976: 152–153) suggested, unemployment does not automatically generate a decline in wages since it depends on the capacity of resistance and organization of workers as an expression of the class struggle; this exogenous factor is not appreciated by most of the scholars. The immediate cause of the recovery of profitability was the emergence of neoliberalism which, according to Harvey (2007: 19, 62), was a “political project to

re-establish the conditions for capital accumulation and to restore the power of economic elites […] dismantling working-class power".

Neoliberalism is the *development way* or *modality*[13] that shows the socio-political and ideological form of assimilation of the new stage of capitalist development. Within the framework of the wasteful and inefficient perception of the State, together with wages, were the cause of inflation during the crisis of 1973–1975[14] (Rodríguez 2005: 63–66), the adoption of the neoliberal way had as its project the reversal of the increase in real wages, a rise that was the solution to the conflict between capital and labor in the Fordist-Keynesian capitalist stage (Ordoñez 2023). The boost to the transnational capitalist class, especially its financial fraction, came with the so-called structural reforms of Ronald Reagan in 1981,[15] which consisted of weakening the Welfare State[16] that granted protection functions for workers, regulated companies, especially those in the financial sector, and increased their spending to create sufficient effective demand (Shaikh 2016: 730; Tsoulfidis and Tsaliki 2022: 12).

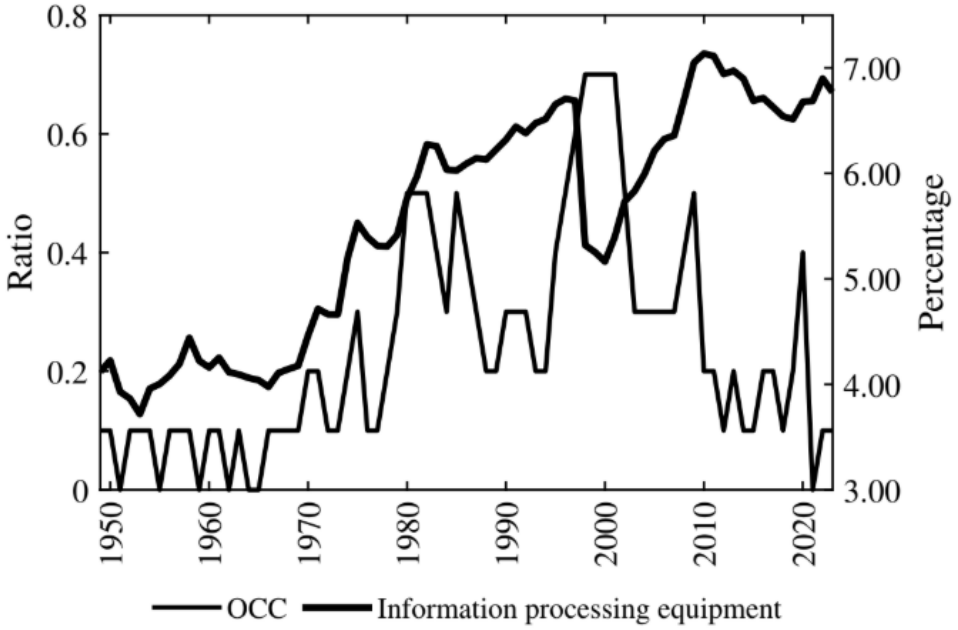
[13]According to the author, the current capitalist stage displays three main alternative ways of development: 1) the privatizing-capitalist way, which is based on the preservation of large transnational corporations and intellectual property as capitalist property; 2) the statist-privatizing-social way, which emphasizes technocratic control of scientific-technical development from "above"; and 3) the popular and democratic-social way, which combines various factors of techno-economic, social and political change "from below" (Dabat 2009: 72–73).

[14] In his annual speech on October 14, 1976, the managing director of the IMF pointed out a change in economic policy to control inflation: "Inflation can only be moderated if fiscal and monetary policies manage to moderate and keep under control the expansion of aggregate demand" (Rodríguez 2005: 4).

[15] During the 1980s, a series of workforce protection laws were reformed, such as the replacement of the Employee Retirement Income Security Act of 1974 by the Retirement Equity Act of 1984 and the Private Pension Reform Act of 1983, which led to the financialization of pensions that disappeared in the 2007–2009 crisis. Likewise, the Social Security Amendments of 1983 were introduced in 1983, which led to the privatization of social security financing (United States Congress 2022).

[16] Following the Great Depression of the 1930s in the United States, a series of reforms were implemented that balanced the balance of power between social classes, known as the New Deal, and embodied in the Welfare State. One of them was the National Labor Relations Act of 1935, which gave the working-class legal mechanisms to improve their working conditions and increase their compensation (The U.S. National Archives and Records Administration 2021).

This *"working-class defeat"* (Mandel 1995: 42) is reflected in the reduction of expenditures such as welfare and social services[17], and wages as a mechanism to control inflation.[18] Because of this, compensation of employees growth rate fell almost vertically in 1981 and since then has maintained an average annual growth of 2 percent, generating the relative stagnation of its share in the national income until the end of 1990, as shown in figure 4. The same figure illustrates the behavior of the rate of surplus value. The first signs of declining profitability appeared at the end of 1960's, although they were short-lived, as recovery began in 1970. This changed in 1982, when it exhibited almost vertical growth.



Sources: U.S. Bureau of Economic Analysis (2025a, 2025b, 2025c, 2025d), and U.S. Bureau of Labor Statistics (2024a).

Figure 4. Growth Rate of the Compensation of Employees and the Rate of profit (1967–2023)

[17] Public spending on welfare and social services peaked at 32 percent in 1970 until reaching around 2 percent in 2019 (U.S. Bureau of Economic Analysis (2024c).

[18] This was derived from the monetarist theory which, according to Hayek (1976: 68 cited in Rodríguez 2005: 113), attributes that "inflation is due solely and exclusively to the monetary expansion arbitrated by our rulers", but the author adds to this cause "the pressures exerted by both the unions and other monopolistic groups such as, for example, the oil cartel". In this way, the struggle by the working class to increase their compensation through their union organization was, in part, the cause of the inflation that led to the crisis of 1973–1975.

Also, it is important to note that the slope of the rate of surplus value during the 1980s was nearly twice that of the 1970s, indicating that profitability was not as high as the next decade, despite the increase in the level of the OCC. These processes that led to the defeat of the working class are considered the immediate cause of the change in the slope of the rate of surplus value observed in the early 1980s. And, by extension, the recovery of the rate of profit, since the VECM suggests that it rises by 18.5 percent for every 100 percent increase in the rate of exploitation.

Hence, the balance of power is tilted towards a greater share of capital gains in the distribution of the product, especially financial capital,[19] as suggested by Duménil and Lévy (2016). This outcome of class struggle was also caused by the deterioration of the wage bargaining power of the working class, which is manifested in the sustained decrease of those affiliated or represented by a union since 1979, as shown in figure 5. During the first five-year period of the 1980s, the growth in union share was negative, falling from −5.1 percent in 1980 to its lowest point, −6.5 percent, in 1984. This has generated wage inequality between unionized and non-unionized workers[20] and between the 0.1 percent of workers with the highest average annual wages and the lowest 90 percent.[21]

[19] Financial profits have been gaining weight in total US corporate profits since the mid-1980s, rising from 5.4 percent in 1984 to 23 percent in 2021 (U.S. Bureau of Economic Analysis 2024 2022a).
[20] This gap went from an average of $1 per hour in 1974 to $2 in 1981, peaking in 2011 at $4.6, and in 2021 it stood at $3.3 (Hirsch and Macpherson 2022).
[21] In 1974, the ratio between the 0.1 percent of workers with the highest average annual wage and the lowest 90 percent was 21, while in 2021 it reached the ratio of 80 (Economic Policy Institute 2025).

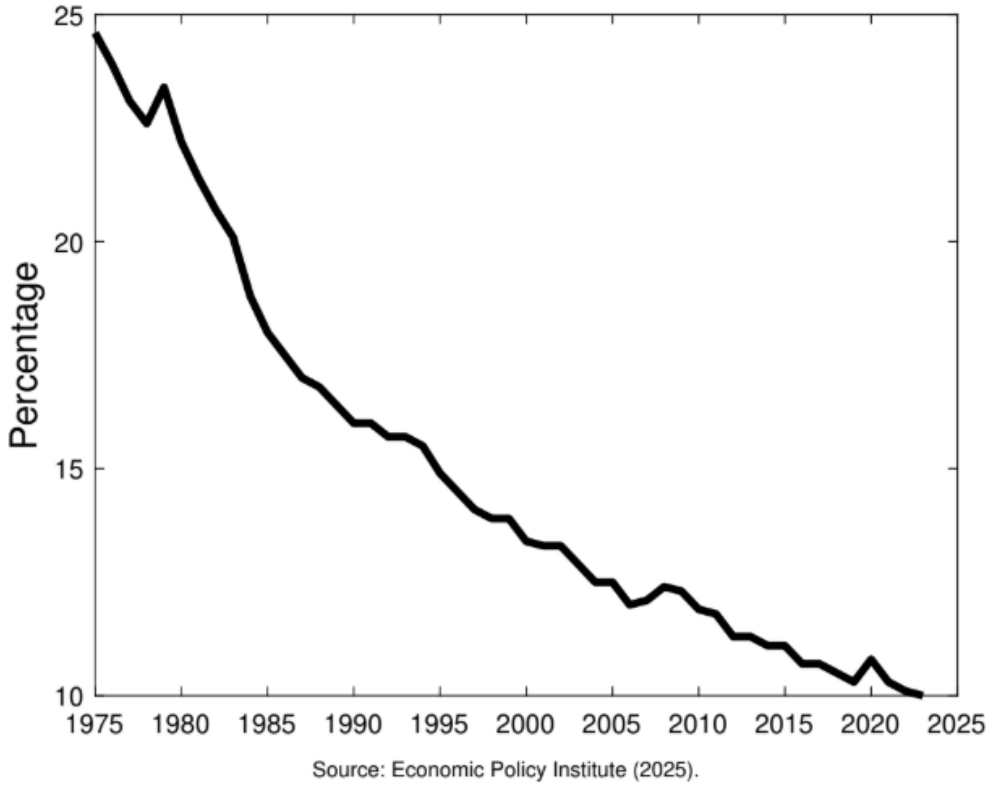


Figure 5. Share of Workers in a Union (1975–2023)

Shaikh (2016, p. 734) notes that the net rate of profit recovered thanks to "cheapened labor and cheapened finance". The causes of this increase were the Fed's monetary policy of lowering interest rates[22] and financial deregulation,[23] Reagan's policies, and "globalization which brought the world's large pool of cheap labor into more direct competition with labor markets in the developed world" (Shaikh, 2016: 730), leading to a growing gap between labor productivity and real wages. Furthermore, the cheapness of capital thanks to the Fed's monetary policy gave an additional boost to the rate of profit to the extent that it allowed the realization of the surplus value

[22] During 1984–2012, one-year and ten-year real interest rates show a downward trend, and even negative in some months (Federal Reserve Bank of Cleveland 2024 2024a).
[23] The rise of the financial sector made it possible to channel the overaccumulated capital from the crises of the 1970s and 1980s into financing the new technology sector.

through consumer credit,[24] which was necessary to counter the decrease in labor compensation (Shaikh 2016: 736). Tsoulfidis and Tsaliki (2022: 12) agree on this, since financial deregulation has led to the reduction of interest rates which, together with the reduction of real wages, restored the rate of profit in the 1980s.

However, for Li et al. (2007: 47), the multiple attacks on the economic and social rights of workers in *neoliberal capitalism*[25] have been a limited success, as they have not meant a substantial impact on wage and tax costs[26], which would explain why the share of wages in the United States and Great Britain during the expansive wave of the fifth long cycle is not significantly lower than the depressive wave of the fourth long cycle, according to their data. It would be, as suggested by Maito (2013: 187), a limited process of *profit squeeze*.

In part, it can be interpreted from Li et al. (2007) that, in the long run, the rate of profit cannot be sustained solely on those exogenous factors driving the rate of surplus value. As Marx pointed out, the rate of profit is fueled by the rise in the OCC, which enables greater surplus value production and, consequently, a higher rate of surplus value. As indicated in table 4, the upward trend in the rate of profit stems both from an increase in the rate of surplus value and from the growth of the OCC through gains in labor productivity. In other words, the 4th technological revolution was essential to sustaining the recovery of the profit rate.

---

[24] From 1984 to 2009, household debt as a percentage of GDP showed an upward trend (from 47.5 to 99 percent) when had remained relatively stable since 1965 (at around 46 percent) (Board of Governors of the Federal Reserve System 2025a).

[25] According to Li et al. (2007: 43), neoliberal capitalism has the following characteristics: "[…] transnational corporations, global mobility of capital and outsourcing, decline of organized labor, weakening of government regulations, monetarist macroeconomic policies […] and the financial expansion of the American hegemony".

[26] However, during the Reagan administration, tax reform was carried out that reduced taxes on big capital through two laws in 1981 and 1986. The Economic Recovery Tax Act of 1981 involved cutting the top personal income tax rate from 70 to 50 percent and the lowest from 14 to 11 percent, expanding the estate tax exemption, reducing taxes on extraordinary profits, and decreasing the capital gains tax from 28 to 20 percent, among other measures (Corporate Finance Institute 2024). The Tax Reform Act of 1986 reduced the top personal income tax rate from 50 to 38.5 percent, decreasing it in subsequent years to 28 percent, and increased the top capital gains tax rate from 20 to 28 percent (Laffer 2004).

On the other hand, it is also understood based on Shaikh (2016: 730) that the *profit squeeze* is not limited to the national economy, but is expanded by the effect of globalization, which puts the American working class in competition with the rest of the world through offshoring and outsourcing processes, meaning extra profits for American transnational corporations. It is the mechanism of export or flow of capital into countries where the average OCC is significantly lower than in the basic industrial branches of the industrialized capitalist countries like the United States, according to Mandel (1995: 11) and Marx (1991, p. 345). This line of research should be pursued in future studies, bearing in mind that the analysis presented here is conducted exclusively from within the US economy.

Up to this point, the evidence suggests that, in qualitative terms, to promote this counteracting influence in the United States, the State was required to assume a corporate character, taking an active role in the transnational inter-capitalist competitive struggle. According to Jean Magniadas (Chesnais et al. 2002: 110), the process of globalization pushed supranational institutions to play a more relevant role in the competition or alliance between nation-states. The incidence of the US government through supranational organizations and with the establishment of international legal frameworks[27] has led to deregulation and open global competition in sectors previously owned or highly regulated by the State in underdeveloped countries (McKinsey Global Institute 2015). Hence, we disagree that the relocation of US corporations leads to the rise of wage and tax costs as the working class is organized in the countries receiving their investment, as suggested by Li et al. (2007: 48).

[27] The United States has 14 free trade agreements signed with 20 countries, bilateral investment treaties with 41 countries, and 34 intellectual property agreements with at least 20 countries (U.S. Department of Commerce 2023).

Furthermore, in its corporate character, the US government directs its investments toward intellectual property products (software and research and development) to drive technological innovation, at the expense of physical investment (infrastructure and equipment).[28] It is the role of the US State in this phase of development to maintain the technological leadership of its corporations through direct participation with public investment, or indirect participation through its fiscal[29] or trade policies[30], within the framework of the global hegemonic struggle, which is based on the construction of a scientific-technological infrastructure and the availability of highly skilled labor. Its realization depends on the dominance of software, which is the focal point of global economic profits (Zhang 2025).

The neoliberal way of capitalist development in the United States would lead to the defeat of the working class through the dismantling of labor protective laws, the deterioration of union organization and bargaining power, and the re-functionalization of the State (an offensive against the Welfare State and its new corporate character). This signified new forms of social relations of production between capital, labor, and the State.

For the above reason, we consider that neoliberalism was the immediate cause of the change in the trend of the rate of surplus value rate as an exogenous cause of the defeat of the American working class through the almost vertical decrease in the growth rate of labor compensations. The change

[28] In recent years, US government investment has shifted towards intellectual property products, which include software and research and development (R&D). In 2002, 61.1 percent of investment was allocated to infrastructure, 25.2 percent to intellectual property, and 16.2 percent to equipment. By 2022, 41.6 percent went to infrastructure, 35.4 percent to intellectual property, and 23.6 percent to equipment (U.S. Bureau of Economic Analysis 2025c).

[29] According to Sherlock and Marples (2022: 13), "historically, the largest component of the federal tax system, in terms of revenue generated, has been the individual income tax". By 2021, 50.5 percent of federal government revenue came from individual income taxes, Social Security or payroll taxes contributed 32.5 percent, while corporate income taxes accounted for 9.2 percent of total revenue. About the earnings generated abroad, "generally, dividends received by US corporate shareholders from their controlled foreign corporations (CFCs) are eligible for a 100 percent deduction on received dividends".

[30] US trade policy is based on liberalization and the enforcement of international trade rules, the promotion of exports, tax-free treatment for developing countries and eligible products, and protection and promotion through investment treaties and trade agreements (Akhtar et al. 2024).

in the trajectory of the rate of surplus value became endogenous to the accumulation process since this recovery of profitability allowed the massive application of the 4th technological revolution. This led to an increase in labor productivity and a cheapening of constant capital, resulting in an increase in the rate of profit from 1984 onward, forming the expansive long wave of the fifth long cycle.

5. Conclusions

The evidence indicates that the US rate of profit rate experienced a regime shift after 1983, inaugurating the expansive wave of a fifth long cycle with a peak in the late 1990s. Methodologically—combining Hodrick–Prescott filter trends (λ=100), Johansen cointegration (VECM), and Bai–Perron breaks—this work shows that profitability's recovery was driven primarily by a steep rise in the rate of surplus value and, secondarily, by a slowdown in OCC growth consistent with productivity gains from the fourth technological revolution. These results reconcile periodization based on Mandel's framework with the recent literature on the US long waves after the 1970s.

These findings imply that the turn to neoliberalism was the immediate catalyst enabling the diffusion of the fourth technological revolution. Then, these new technologies sustained profitability through labor productivity until it peaked in the late 1990s. The persistent post-mortgage economic crisis weakness suggests that the expansive wave has long since ended, giving way to a depressive wave punctuated by structural breaks (2007, 2021). Future research should further examine if the fourth technological revolution has not yet run its course since the current state of the US long cycle, as well as integrate the geopolitical and economic transformations that followed the COVID-19 pandemic into this long-term analysis.

6. Appendix

References

Akhtar, Shayerah I., Cimino-Isaacs, Cathleen D., and Sutter, Karen M. 2024. *U.S. trade policy: Background and current issues*. Congressional Research Service (CRS). Accessed at: https://crsreports.congress.gov/product/pdf/IF/IF10156

Basu, Deepankar. 2016. *Long waves of capitalist development: An empirical investigation* Department of Economics, University of Massachusetts Working Paper 2016-15. Amherst: University of Massachusetts. Accessed at: https://www.econstor.eu/handle/10419/174407

Board of Governors of the Federal Reserve System. 2025. *60-Day AA nonfinancial commercial paper interest rate [CPN2M]*. FRED, Federal Reserve Bank of St. Louis. Accessed at: https://fred.stlouisfed.org/series/CPN2M

Board of Governors of the Federal Reserve System. 2025a. *Financial soundness indicator, households; debt as a percent of gross domestic product, level*. FRED, Federal Reserve Bank of St. Louis. Accessed at: https://fred.stlouisfed.org/series/BOGZ1FL010000336A

Board of Governors of the Federal Reserve System. 2025b. *Industrial production: Manufacturing (SIC) [IPMANSICS]*. FRED, Federal Reserve Bank of St. Louis. Accessed at: https://fred.stlouisfed.org/series/IPMANSICS

Bolt, Jutta, and van Zanden, Jan Luiten. 2024. Maddison style estimates of the evolution of the world economy: A new 2023 update. *Journal of Economic Surveys* Special Issue: 631–671. Accessed at: https://doi.org/10.1111/joes.12618

Castells, Manuel. 2022. The network society revisited. *American Behavioral Scientist* 67 (7): 940 –946. Accessed at: https://doi.org/10.1177/00027642221092803

Chesnais, François, Duménil, Gérard, Husson, Michel, Lévy, Dominique, Magniadas, Jean, Samary, Catherine, Serfati, Claude, and Wallerstein, Immanuel. 2002. Discusión. ¿Una nueva fase del capitalismo? Tres interpretaciones marxistas. In *La globalización y sus crisis: Interpretaciones desde la economía crítica*. Madrid: Los Libros de la Catarata.

Corporate Finance Institute. 2024. *Economic Recovery Tax Act of 1981 (ERTA)*. Accessed at: https://corporatefinanceinstitute.com/resources/economics/economic-recovery-tax-act-of-1981-erta/

Dabat, Alejandro. 2009. Economía del conocimiento y capitalismo informático. Notas sobre estructura, dinámica y perspectivas de desarrollo. In *Globalización, conocimiento y desarrollo. La nueva economía del conocimiento. Estructura y problemas. Tomo I.*, eds. Alejandro Dabat, and José de Jesús Rodríguez Vargas. Ciudad de México: Instituto de Investigaciones Económicas, Universidad Nacional Autónoma de México.

Duménil, Gérard, and Lévy, Dominique. 2016. Technology and distribution in managerial capitalism: The chain of historical trajectories à la Marx and countertendential traverses. *Science & Society* 80 (4): 530–549. Accessed at: https://doi.org/10.1521/siso.2016.80.4.530

Economic Policy Institute. 2025. *Historical union membership*. Accessed at: https://www.epi.org/data/

Economic Policy Institute. (2025b). *Wages for top 1.0%, 0.1%, and bottom 90%*. Accessed at: https://www.epi.org/data/

Federal Reserve Bank of Cleveland. (2024). *1-year real interest rate*. FRED, Federal Reserve Bank of St. Louis. Accessed at: https://fred.stlouisfed.org/series/REAINTRATREARAT1YE

Federal Reserve Bank of Cleveland. (2024a). *10-year real interest rate*. FRED, Federal Reserve Bank of St. Louis. Accessed at: https://fred.stlouisfed.org/series/REAINTRATREARAT10Y

Harvey, David. 1992. *The condition of postmodernity: An enquiry into the origins of cultural change*. Cambridge: Blackwell Publishers.

Harvey, David. 2007. *A brief history of neoliberalism*. New York: Oxford University Press.

Hirsch, Barry T., and Macpherson, David A. 2022. II. Historical tables: Union wages, nonunion wages, and union wage gaps, 1973–2021. *Union Membership and Coverage Database*. Accessed at: https://www.unionstats.com/

Laffer, Arthur B. 2004. The Laffer curve: Past, present, and future. *The Heritage Foundation*. Accessed at: https://www.heritage.org/taxes/report/the-laffer-curve-past-present-and-future

Li, Minqi, Xiao, Feng, and Zhu, Andong. 2007. Long waves, institutional changes, and historical trends: A study of the long-term movement of the profit rate in the capitalist world-economy. *Journal of World-Systems Research* 13 (1): 33–54. Accessed at: https://doi.org/10.5195/jwsr.2007.360

Louçã, Francisco. 1999. Nikolai Kondratiev and the early consensus and dissensions about history and statistics. *History of Political Economy* 31 (1): 169–205. Accessed at: https://doi.org/10.1215/00182702-31-1-169

Maito, Esteban Ezequiel. 2013. Distribución del ingreso, rotación del capital y niveles de rentabilidad. *Economía: teoría y práctica* (45): 169-206.

Mandel, Ernest. 1976. *Late capitalism*. London: NLB.

Mandel, Ernest. 1981. Explaining long waves of capitalist development. *Futures* 13 (4): 332-338. Accessed at: https://doi.org/10.1016/0016-3287(81)90148-8

Mandel, Ernest. 1992. Introduction. En K. Marx, *Capital: Volume 2.* London: Penguin Books.

Mandel, Ernest. 1995. *Long waves of capitalist development: A Marxist interpretation*. New York: Verso Books.

Marx, Karl. 1982. *Capital: Volume 1*. London: Penguin Books.

Marx, Karl. 1991. *Capital: Volume 3*. London: Penguin Books.

McKinsey Global Institute. 2015. *Playing to win: The new global competition for corporate profits*. McKinsey & Company. Accessed at: https://www.mckinsey.com

National Bureau of Economic Research. (2025). *Index of manufacturing production for United States [A0107AUSA322NNBR]*. FRED, Federal Reserve Bank of St. Louis. Accessed at: https://fred.stlouisfed.org/series/A0107AUSA322NNBR

National Bureau of Economic Research. (2025a). *Index of the general price level for United States [M04051USM324NNBR]*. FRED, Federal Reserve Bank of St. Louis. Accessed at: https://fred.stlouisfed.org/series/M04051USM324NNBR

National Bureau of Economic Research. (2025b). *Total exports for United States [M07023USM144NNBR]*. FRED, Federal Reserve Bank of St. Louis. Accessed at: https://fred.stlouisfed.org/series/M07023USM144NNBR

Ordóñez, S. (2023). New phase of development and knowledge capitalism: Gramsci's historical revenge? *International Gramsci Journal* 1 (2). Accessed at: https://doi.org/10.14276/igj.v1i2.3986

Ordóñez, S. (2024). Global production networks and dynamic cores in the world's main nodes: The technological-productive transition of the automotive industry. *World Review of Political Economy* 15 (1): 46–81. Accessed at: https://www.jstor.org/stable/48774392

Organization for Economic Co-operation and Development. (2025). *Interest rates: Long-term government bond yields: Combined terms: Total for United States [IRLTCT01USQ156N]*. FRED, Federal Reserve Bank of St. Louis. Accessed at: https://fred.stlouisfed.org/series/IRLTCT01USQ156N

Roberts, Michael. (2016). *The long depression: How it happened, why it happened, and what happens next*. Chicago: Haymarket Books.

Rodríguez Vargas, José de Jesús. 2005. *La nueva fase de desarrollo económico y social del capitalismo mundial.* PhD dissertation. Ciudad de México: Facultad de Economía, Universidad Nacional Autónoma de México. Accessed at: https://ru.dgb.unam.mx/items/a1e439f7-69c3-41a2-b041-0cc95a4832db

Shaikh, Anwar. 2016. *Capitalism: Competition, conflict, crises*. New York: Oxford University Press.

Shaikh, Anwar, and Tonak, E. Ahmet. 2007. *Measuring the wealth of nations: The political economy of national accounts*. New York: Cambridge University Press.

Sherlock, Molly F., and Marples, Donald J. 2022. *Overview of the federal tax system in 2022*. Congressional Research Service (CRS). Accessed at: https://crsreports.congress.gov/product/pdf/R/R45145

Shiller, Robert J. 2015. *Irrational exuberance*. Princeton: Princeton University Press.

Smits, J. P., Woltjer, Pieter J., and Ma, Debin. 2009. A dataset on comparative historical national accounts, ca. 1870–1950: A time-series perspective. *Groningen Growth and Development Centre Research Memorandum GD-107*. University of Groningen.

Snyder, Hannah. 2019. Literature review as a research methodology: An overview and guidelines. *Journal of Business Research* 104: 333–339. Accessed at: https://doi.org/10.1016/j.jbusres.2019.07.039

The U.S. National Archives and Records Administration. (2021). *National Labor Relations Act (1935)*. Accessed at: https://www.archives.gov/milestone-documents/national-labor-relations-act

Tsoulfidis, Lefteris, and Tsaliki, Persefoni. 2022. The long recession and economic consequences of the Covid-19 pandemic. *Investigación Económica* 81 (321): 3–29. https://www.jstor.org/stable/48676880

United Nations Industrial Development Organization. (2019). *Informe sobre el desarrollo industrial 2020: La industrialización en la era digital*. Accessed at: https://ods9.org/resource/136/informe-sobre-el-desarrollo-industrial-2020

United States Congress. (2022). *Legislation*. Congress.gov | Library of Congress. Accessed at: http://www.congress.gov/

U.S. Bureau of Economic Analysis. (2024). *Corporate profits with inventory valuation adjustments: Domestic industries: Financial [A392RC1Q027SBEA]*. FRED, Federal Reserve Bank of St. Louis. Accessed at: https://fred.stlouisfed.org/series/A392RC1Q027SBEA

U.S. Bureau of Economic Analysis. (2024a). *Corporate profits with inventory valuation and capital consumption adjustments: Domestic industries: Nonfinancial [A463RC1Q027SBEA]*. FRED, Federal Reserve Bank of St. Louis. Accessed at: https://fred.stlouisfed.org/series/A463RC1Q027SBEA

U.S. Bureau of Economic Analysis. (2024b). *Glossary*. U.S. Bureau of Economic Analysis (BEA). Accessed at: https://www.bea.gov/help/glossary?page=5

U.S. Bureau of Economic Analysis. (2024c). *Government current expenditures: Income security: Welfare and social services [G160371A027NBEA]*. FRED, Federal Reserve Bank of St. Louis. Accessed at: https://fred.stlouisfed.org/series/G160371A027NBEA

U.S. Bureau of Economic Analysis. (2025). *Exports of goods and services [EXPGSA]*. FRED, Federal Reserve Bank of St. Louis. Accessed at: https://fred.stlouisfed.org/series/EXPGSA

U.S. Bureau of Economic Analysis. (2025a). *Fixed assets accounts tables*. U.S. Bureau of Economic Analysis (BEA). Accessed at: https://www.bea.gov/data/economic-accounts/national

U.S. Bureau of Economic Analysis. (2025b). *Industry economic accounts data*. U.S. Bureau of Economic Analysis (BEA). Accessed at: https://www.bea.gov/data/gdp/gdp-industry

U.S. Bureau of Economic Analysis. (2025c). *National income and product accounts*. U.S. Bureau of Economic Analysis (BEA). Accessed at: https://www.bea.gov/data/economic-accounts/national

U.S. Bureau of Economic Analysis. (2025d). *National income: Compensation of employees [A033RC1A027NBEA]*. FRED, Federal Reserve Bank of St. Louis. Accessed at: https://fred.stlouisfed.org/series/A033RC1A027NBEA

U.S. Bureau of Labor Statistics. (2024). *Business sector: Labor productivity (output per hour) for all workers [PRS84006091]*. FRED, Federal Reserve Bank of St. Louis. Accessed at: https://fred.stlouisfed.org/series/PRS84006091

U.S. Bureau of Labor Statistics. (2024a). *Contribution of information processing equipment (IPE) intensity to labor productivity*. U.S. Bureau of Labor Statistics. Accessed at: https://www.bls.gov/productivity/

U.S. Bureau of Labor Statistics. (2024b). *Glossary*. U.S. Bureau of Labor Statistics. Accessed at: https://www.bls.gov/bls/glossary.htm

U.S. Bureau of Labor Statistics. (2024c). *Productivity glossary*. U.S. Bureau of Labor Statistics. Accessed at: https://www.bls.gov/productivity/glossary.htm

U.S. Bureau of Labor Statistics. (2024d). *Unemployment rate [UNRATE]*. FRED, Federal Reserve Bank of St. Louis. Accessed at: https://fred.stlouisfed.org/series/UNRATE

U.S. Department of Commerce. (2023). *Trade agreements*. International Trade Administration. Accessed at: https://www.trade.gov/trade-agreements

USA Government. (2022). *TAC Program - Making trade agreements work for you!* Official Guide to Government Information and Services. Accessed at: https://web.archive.org/web/20170708210813/http://tcc.export.gov/Trade_Agreements/Intellectual_Property_Rights/index.asp

World Bank. (2024). *GDP (constant 2015 US$)*. World Bank Open Data. Accessed at: https://data.worldbank.org

World Bank. (2024a). *GDP, PPP (constant 2021 international $)*. World Bank Open Data. Accessed at: https://data.worldbank.org

Zhang, Kevin Honglin. (2025). Industrial policy in the competition between an existing hegemon and a rising superpower: The case of the US and China. *The Chinese Economy*: 1–16. https://doi.org/10.1080/10971475.2025.2469198